

CAMERON LEWIS

ELECTRICAL ENGINEERING STUDENT

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SUMMARY

Electrical Engineering student at Memorial University with hands-on industry experience in PCB design, schematic capture, and hardware bring-up. Proven ability to perform design verification and technical documentation in fast-paced, multidisciplinary environments, with a strong foundation in analog and digital circuit design and signal integrity.

TECHNICAL SKILLS

PCB Design	Altium, KiCad, schematic capture, layout, wiring harness design, PDM
Circuit Design	Analog and digital circuits, power electronics, DC-DC converter design, signal integrity
Simulation & Test	LTspice, PSpice, oscilloscope, logic analyzer, hardware bring-up, soldering
Embedded Systems	C, C++, Python, MATLAB, Arduino IDE, microcontroller integration
Tools & Workflow	Git, Cadence Allegro (exposure), technical documentation, cross-disciplinary collaboration

PROFESSIONAL EXPERIENCE

Electrical Engineering Student - Co-op

January 2026 – April 2026

Kraken Robotics Systems Inc. - Mount Pearl, NL

- Designed PCBs for subsea systems including Synthetic Aperture Sonar and Katfish ROTV, applying signal integrity best practices to meet manufacturability requirements across multiple product lines.
- Redesigned wiring harness layouts by incorporating signal integrity analysis and direct engineer feedback, reducing potential failure points in cable assemblies.
- Maintained technical documentation for 5+ products using PDM software, enabling accurate version control and streamlining design review and verification approvals.

Electrical Team Member

September 2025 – Present

Paradigm Engineering - MUN Student Design Team - St. John's, NL

- Designed multilayer control PCBs integrating microcontrollers, power-management circuits, and voltage regulators using KiCad for an autonomous competition kart, verifying functionality with oscilloscope measurements.
- Contributed to the design and assembly of the Auxiliary Control Board PCB, managing power distribution with LDOs, level shifters, and protection circuitry, and validated operating margins across electrical subsystems.

PROJECTS

Brick Buck Board

April 2026 – May 2026

- Designed a custom power electronics PCB converting a 48 V battery input into regulated 12 V and 5 V rails for autonomous kart electronics, incorporating input protection, output filtering, and power-status indicators.
- Verified voltage rail performance and operating margins using oscilloscope measurements to validate design robustness prior to integration.

Auxiliary Control Board (Paradigm)

April 2026

- Designed a custom auxiliary control PCB managing voltage regulation, level shifting, protection circuitry, and power distribution across multiple subsystems of the Paradigm autonomous kart.
- Conducted hardware bring-up and debug using oscilloscope testing to verify signal behaviour and confirm compliance with design specifications.

Thermal Camera

February 2026 – Present

- Developed an ESP32-based thermal imaging system using an infrared sensor array to output a real-time heat map, integrating sensor acquisition, display output, and power management into a portable embedded platform.

EDUCATION

Bachelor of Engineering (Co-op) - Electrical Engineering

2024 – Present

Memorial University of Newfoundland · Class of 2029 · GPA 3.7 · Dean's List 2024–2025

High School Diploma

2018 – 2024

Roncalli Central High School · Salutatorian · GPA 4.0 · Teachers' Academic Award

ACHIEVEMENTS & INTERESTS

Autonomous Karting Series 2026 - Lead Control Module Designer

May 2026

Attended the Autonomous Karting Series 2026 in Indiana, USA, hosted by Purdue University, as the lead designer for the kart's control module.

Table Tennis - Team NL

May 2025

Represented Newfoundland and Labrador at the Atlantic Championship Tournament in Halifax, Nova Scotia, competing against all Atlantic provinces.